

NARWC Database Reference

Sources: NARWC Reference Document 2023-01, Version 8 (R.D. Kenney, URI/GSO); *src/+narwc/+db/FieldDefinitions.m*; validation rule modules in *src/+narwc/+validation/+rules/*; lookup table snapshots in *data/tables/*.

1. Database Overview

The North Atlantic Right Whale Consortium (NARWC) Sightings Database archives cetacean, sea turtle, and marine mammal sightings from aerial and shipboard surveys and opportunistic reports conducted in the western North Atlantic Ocean from the late 1970s to the present. The database contains approximately 10.8 million event records spanning systematic dedicated surveys, platforms-of-opportunity programs, citizen science reports, and historical literature records.

1.1 History and Lineage

The database structure originates with the Cetacean and Turtle Assessment Program (CETAP), a large-scale systematic aerial and shipboard survey of the northeastern U.S. continental shelf conducted from 1978 to 1982 under contract with the U.S. Bureau of Land Management. When the NARWC was established in the mid-1980s and began consolidating right whale sighting data from multiple contributors, the CETAP database schema was adopted as the foundation. Legacy CETAP codes, field names, and conventions persist throughout the documentation and some lookup tables.

The database was originally implemented in dBASE and analyzed with SAS macros written at URI/GSO. A migration to SQL Server has since been completed. The MATLAB codebase on the refactor branch provides import, validation, and processing tools for the SQL Server database.

The current authoritative documentation is the NARWC Reference Document, maintained by R.D. Kenney (URI/GSO). This document synthesizes Version 8 (Reference Document 2023-01). Where the PDF and the codebase conflict, Section 6 records the discrepancy rather than resolving it unilaterally; the code is generally the ground truth for what is currently enforced.

1.2 Data Categories

Category	Description
Dedicated aerial	Systematic line-transect surveys flown by fixed-wing aircraft or helicopter
Shipboard / POP	Systematic surveys from vessels; includes Platforms of Opportunity programs
Intermediate format	Data submitted in a standardized intermediate format (e.g., CCS 2015+, MassCEC)
Opportunistic	Ad hoc sightings from vessels, shore stations, other aircraft, citizen science, and published literature

Each row in the database corresponds to a single event: a sighting, a non-sighting weather or effort record, a leg boundary marker, or a watch transition.

1.3 Version 8 Changes

Version 8 of the reference document (2023) introduced or formalized:

- ANGLEL and ANGLER: declination angles to sightings from the trackline, replacing STRIP for NEAQ aerial surveys beginning in 2022.
- New SPECCODE values: CV-C (construction barge – crane), CV-O (other construction), CV-P (pile-driver), CV-R (rock dumper), CREW (offshore wind support vessel), CABL (cable/pipe laying, reinstated after earlier discontinuation).
- New BEHAV codes 73 and 74: actively fishing, and actively fishing with deployed gear (for vessel sightings).
- ANHEAD code 20: 360° milling or circling (no consistent heading).
- TAXCODE=0: formalized for vessels, gear, human activities, and debris/pollution sightings.

2. Survey Types and FILEID Conventions

2.1 FILEID Structure

FILEID is an 8-character alphanumeric identifier unique to each submitted data file. The encoding:

Position	Content
1	Survey type code (uppercase = pre-2000; lowercase = 2000 and later)
2	Contributor or program identifier
3–4	Two-digit year of survey
5–7	Julian date (day of year) of the first data day in the file

Position	Content
8	Sequence or qualifier code

The case convention on position 1 is a date marker only: uppercase indicates the file contains data from before 2000; lowercase indicates 2000 or later. This applies to position 1 only.

2.2 Survey Type Codes (Position 1)

Code (pre-2000 / 2000+)	Type
A / a	Dedicated aerial line-transect survey
F / f	Shipboard survey (POP and similar)
H / h	Intermediate format (e.g., CCS, MassCEC)
O / o	Opportunistic sightings

The lookup table `data/tables/DType.csv` (5 rows) records the currently active survey type codes. The fifth row has not been fully documented in the available PDF content; see Section 6.4.

3. Field Reference

All 55 fields currently defined in `narwc.db.FieldDefinitions` are described below, grouped by function. Fields that appear in the PDF but are absent from `FieldDefinitions` are listed in Section 7.

Where a field is constrained by a lookup table, the table filename and current row count are given. FK validation against lookup tables is handled by `foreign_key_rules.m` and by field-specific rule modules.

3.1 Survey Identity and Structure

FILEID

Type	string
NULL	Not allowed
Lookup	—
Surveys	All

8-character survey file identifier (see Section 2). All event records within a single submitted file share the same FILEID. Listed as required in `required_fields.m` (that rule file carries a FIXME noting the required-field list is incomplete).

EVENTNO

Type	double
NULL	Not allowed
Lookup	—
Surveys	All

Sequential event number within a file, assigned by the data-logging program (Logger, D-Tracker, Mysticetus) or manually. Each keypress creating a record increments EVENTNO. Must be unique within a file; gaps are acceptable. Listed as required in `required_fields.m`.

SIGHTNO

Type	double
NULL	Required for sightings; must be absent for non-sightings
Lookup	—
Surveys	All (sighting records only)

Sighting number, sequential from 1 within a file. Required for all sighting records and prohibited for non-sighting records. Duplicate SIGHTNOs within a file are not allowed; gaps are acceptable. During CETAP, SIGHTNO=999 was assigned to non-target species (seals, sharks, sunfish) to facilitate removal. Datasets submitted in dBASE format use SIGHTNO=0 for non-sightings; these are converted to true NULLs during import.

The SIGHTNO/non-sighting mutual-exclusion rule is described in the PDF but is not enforced by any current rule module; see Section 6.2.

BLOCK

Type	string
NULL	Allowed
Lookup	<code>data/tables/Block.csv</code> (55 rows)
Surveys	Dedicated aerial; some shipboard

Alphanumeric code identifying the geographic survey block within a stratified or systematic survey design.

STRATUM

Type	string
NULL	Allowed
Lookup	data/tables/STRATUM.csv (10 rows)
Surveys	Dedicated aerial (stratified)

One-character depth stratum or sub-block identifier. Valid values:

Code	Meaning
X	0–20 fathoms
Y	20–50 fathoms
Z	> 50 fathoms
0	Non-stratified aerial survey block
A, B	Scotian Shelf block halves (1987+)
I	Florida, inshore (1989–1992 MMS surveys)
O	Florida, offshore
M	NLPSC year 2+, Martha's Vineyard area
R	NLPSC year 2+, Rhode Island (expanded area, 2012+)

LEGNO

Type	double
NULL	Allowed
Lookup	—
Surveys	Dedicated aerial and shipboard

Sequential leg number within a file. Increments each time a new effort leg begins. Gaps are permitted.

LEGTYPE

Type	double
NULL	Allowed
Lookup	data/tables/LEGTYPE.csv (9 rows)
Surveys	Dedicated aerial and shipboard

Code for the structural type of a leg (transect, transit, circling, etc.). LEGTYPE=2 designates an on-effort transect leg. Combined with LEGSTAGE=2, it defines the “on-effort” condition that governs STRIP requirements.

LEGSTAGE

Type	double
NULL	Allowed
Lookup	data/tables/LEGSTAGE.csv (9 rows)
Surveys	Dedicated aerial and shipboard

Code for the stage within a leg (e.g., beginning of watch, on watch, end of watch). LEGSTAGE=2 indicates observers are actively on watch. Together with LEGTYPE=2, defines on-effort status.

DDSOURCE

Type	string
NULL	Not allowed
Lookup	data/tables/DDSOURCE.csv (48 rows)
Surveys	All

Data delivery source code, identifying the program or contributor that submitted the data. Listed as required in `required_fields.m`.

IDSOURCE

Type	string
NULL	Not allowed
Lookup	data/tables/IDSOURCE.csv (53 rows)
Surveys	All

Identification source code, recording who or what made the species identification. Listed as required in `required_fields.m`.

3.2 Date and Time

YEAR

Type	double
NULL	Not allowed
Lookup	—
Surveys	All

Four-digit calendar year. Required for all records. Formerly stored as two digits; expanded to four digits for Y2K. `datetime_rules.m` enforces range 1900–(current year + 1) as an error (`validation.datetime.year_min/year_max`) and issues a warning for values before 1980 (`validation.datetime.year_warning`).

MONTH

Type	double
NULL	Allowed
Lookup	data/tables/MONTH.csv (16 rows)
Surveys	All

Calendar month (1–12). `datetime_rules.m` enforces $1 \leq \text{MONTH} \leq 12$. The MONTH.csv lookup has 16 rows; the extra rows are for the seasons (winter, spring, summer, fall)

DAY

Type	double
NULL	Allowed
Lookup	—
Surveys	All

Day of month (1–31). `datetime_rules.m` enforces $1 \leq \text{DAY} \leq 31$ and validates YEAR/MONTH/DAY as a legal calendar date.

TIME

Type	double
NULL	Allowed (opportunistic records only)
Lookup	—
Surveys	All (required for aerial and shipboard; optional for opportunistic)

Event clock time in HHMMSS format, 24-hour UTC. During a 2020 database update all archived times were converted from Eastern Standard Time to UTC. Legacy four-digit times (hhmm) are expanded to HHMMSS by appending “00” during archival; four-digit times are no longer accepted for new survey submissions.

`datetime_rules.m` validates $\text{HH} < 24$, $\text{MM} < 60$, $\text{SS} < 60$, and the full value < 240000 .

3.3 Position and Geometry

LAT_DD

Type	double
NULL	Required for sightings; allowed for some event types
Lookup	—
Surveys	All

Latitude of the survey platform at the moment of the event, in decimal degrees north (positive = north). `coordinate_rules.m` enforces $-90 \leq \text{LAT_DD} \leq 90$ (error) and issues a warning for values outside the configured survey-area bounds (default 20–55°N, `validation.coordinates.study_area_lat_min/max`). LAT_DD and LONG_DD must both be present or both absent.

LONG_DD

Type	double
NULL	Required for sightings; allowed for some event types
Lookup	—
Surveys	All

Longitude of the survey platform at the moment of the event, in decimal degrees (negative = west). Western Atlantic survey data will have negative values. `coordinate_rules.m` enforces $-180 \leq \text{LONG_DD} \leq 180$ (error) and issues a warning for values outside the configured survey-area bounds (default -85 to -40°W, `validation.coordinates.study_area_lon_min/max`).

ALT

Type	double
NULL	Allowed (non-aerial records)
Lookup	—
Surveys	Aerial only

Flight altitude. Per the reference PDF, values are in feet. The description in `FieldDefinitions` (“Altitude in meters”) is incorrect. No range validation is currently implemented.

HEADING

Type	double
NULL	Allowed
Lookup	—
Surveys	All

Compass heading of the survey platform in degrees (0–360). Not to be confused with ANHEAD, which records the observed animal's heading. No validation is currently implemented for this field.

STRIP

Type	double
NULL	Required for on-effort animal sightings (LEGTYPE=2, LEGSTAGE=2); not allowed otherwise
Lookup	data/tables/STRIP.csv (16 rows)
Surveys	Aerial line-transect

Two-digit right-angle distance interval of the sighting from the survey trackline. Odd numbers indicate the left (port) side; even numbers indicate right (starboard). Code 0 (on trackline) applies only to the AT-11 aircraft.

CETAP and early NLPSC codes (1978–2011):

Codes	Interval
0	Trackline (AT-11 only)
1, 2	0– $\frac{1}{4}$ n.mi.
3, 4	0– $\frac{1}{8}$ n.mi.
5, 6	$\frac{1}{8}$ – $\frac{1}{4}$ n.mi.
7, 8	$\frac{1}{4}$ – $\frac{1}{2}$ n.mi.
9, 10	$\frac{1}{2}$ – $\frac{3}{4}$ n.mi.
11, 12	$\frac{3}{4}$ –1 n.mi.
13, 14	>1 n.mi. (AT-11) / 1–2 n.mi. (Skymaster)
15, 16	>2 n.mi. (Skymaster)

NLPSC/MassCEC Skymaster codes (October 2011 onward):

Codes	Interval
1, 2	$<\frac{1}{8}$ n.mi.
3, 4	$\frac{1}{8}$ – $\frac{1}{4}$ n.mi.
5, 6	$\frac{1}{4}$ – $\frac{1}{2}$ n.mi.

Codes	Interval
7, 8	½–1 n.mi.
9, 10	1–2 n.mi.
11, 12	2–4 n.mi.
13, 14	>4 n.mi.

In 2022 the NEAQ aerial survey team stopped recording STRIP and switched to declination angles (ANGLEL, ANGLER). The STRIP requirement for on-effort sightings is documented in the PDF but is not enforced by any current rule module; see Section 6.2.

ANGLEL

Type	double
NULL	Allowed
Lookup	—
Surveys	Aerial line-transect (NEAQ, 2022+)

Declination angle (degrees below horizontal) to a sighting on the left (port) side of the trackline. Added in Version 8 to replace STRIP for the NEAQ aerial program. Provides more precise right-angle distances than strip codes and is less sensitive to changes in survey altitude.

ANGLER

Type	double
NULL	Allowed
Lookup	—
Surveys	Aerial line-transect (NEAQ, 2022+)

Declination angle to a sighting on the right (starboard) side of the trackline. Companion to ANGLEL. Exactly one of ANGLEL/ANGLER should be populated for a given sighting; no rule currently enforces mutual exclusivity.

S_LAT

Type	double
NULL	Allowed
Lookup	—
Surveys	Aerial line-transect (NLPSC/MassCEC 2011+)

Latitude of the exact position of the sighting (not the platform) in decimal degrees north. Added for the 2011–12 NLPSC/MassCEC aerial surveys to support trigonometric calculation of right-angle distances and to verify STRIP estimates. Analogous to the retired CETAP fields ALATDEG/ALATMIN/ALATSEC. S_LAT and S_LONG should both be present or both absent (not currently enforced).

S_LONG

Type	double
NULL	Allowed
Lookup	—
Surveys	Aerial line-transect (NLPSC/MassCEC 2011+)

Longitude of the exact sighting position in decimal degrees (negative = west). Companion to S_LAT. Analogous to the retired CETAP fields ALONDEG/ALONMIN/ALONSEC.

S_TIME

Type	double
NULL	Allowed
Lookup	—
Surveys	Aerial line-transect (NLPSC/MassCEC 2012+)

Clock time (HHMMSS, UTC) at which the exact-position sighting (S_LAT/S_LONG) was recorded during circling. Added after the first year of NLPSC surveys to assess elapsed time between initial trackline detection and the circling position. May differ from TIME, which records the initial detection moment.

3.4 Species and Count

SPECCODE

Type	string
NULL	Required for sightings; must be blank for non-sightings
Lookup	data/tables/SPECCODE.csv (317 rows)
Surveys	All (sighting records only)

Four-letter species or category code. Required for all sighting records and prohibited for non-sighting records. Contributors may not create new codes; additions require approval from the database manager. The legacy dBASE field name is CETSPPCD.

SPECCODE.csv contains all 317 active codes. Key codes by taxonomic group:

TAXCODE=1 (large cetaceans): BLWH, BOWH, BRWH, FIWH, GRWH, HUWH, RIWH (North Atlantic right whale), SEWH, SPWH, SRWH, UNBA, UNBS, UNFS, UNLW, UNRO, UNWH

TAXCODE=2 (medium cetaceans): BEWH, BLBW, GEBW, GOBW, KIWH, MIWH, NBWH, SOBW, TRBW, UNBW, UNMW

TAXCODE=3 (small cetaceans): ASDO, BELU, BODO, CLDO, DSWH, FKWH, FRDO, GRAM, HAPO, LFPW, MHWH, NARW (narwhal), OBDO, PIWH, PSDO, PSWH, PYKW, RTDO, SADO, SFPW, SNDO, SPDO, STDO, TADO, UNBD, UNBF, UNCW, UNDO, UNGD, UNKO, UNLD, UNSB, UNST, WBDO, WSDO

TAXCODE=4 (other marine mammals): BESE, GRSE, HASE, HGSE, HOSE, HPSE, MANA, PINN, POBE, RISE, UNCE, UNMM, UNSE, WALR

TAXCODE=5 (sea turtles): GRTU, HATU, LETU, LOTU, ORTU, RITU, UNTU

TAXCODE=6 (sharks): ANSH, BASH, BLSH, DUSH, GHSH, HHSH, LMSH, MKSH, SDOG, SMSH, THSH, TISH, UNSH, WSHH, WTSB

TAXCODE=7 (other fish): BFTU, BLFI, CDRA, CNRA, FLFI, MAHI, MARA, MOBU, OCSU, OTBI, SCFI, SCRA, SWFI, TUNS, UNFI, UNRA, WHMA, YFTU

TAXCODE=8 (birds): ~50+ codes; see SPECCODE.csv. Includes NOGA, GRSH, NOFU, HERG, GBBG, ATPU, COMU, and many others.

TAXCODE=9 (other): AMAL, JELL, LMJE, PMOW, UNID, ZOOP

TAXCODE=0 (vessels, gear, human activity, debris): Active codes include CABL, CG-B, CG-C, CG-S, CG-U, CREW, CRSH, CV-C, CV-O, CV-P, CV-R, DE-B through DE-W, DR-D through DR-W, ECOT, EXPL, FE-H/S/U, FG-A through FG-U, FRNT, FV-C through FV-Z, HELO, JETS, KAYK, LE-V, METT, MV-B through MV-U, MY-L/S, NV-L/S/U, OI-D/L/P/S, OW-B, PIBO, RECV, RV-G/L/S/U/W, SONO, SPFV, SV-L/S/U, UNVE, WHAL, and others. Italicized CETAP codes (AC-J, AC-P, AC-S, AC-T, BT-H, BT-L, BT-M, DIVE, DU-G, DU-T, MULT, SONR, SWIM) are obsolete and must not be used.

species_rules.m validates SPECCODE values against SPECCODE.csv.

TAXCODE

Type	double
NULL	Required for sightings
Lookup	data/tables/TAXCODE.csv (10 rows)

Surveys	All (sighting records only)
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One-digit taxonomic category. Values: 0=vessel/gear/human activity/debris, 1=large cetacean, 2=medium cetacean, 3=small cetacean, 4=other marine mammal, 5=sea turtle, 6=shark, 7=other fish, 8=bird, 9=other/unknown.

Per the reference PDF, TAXCODE is assigned automatically by a SAS macro during data entry at GSO and is therefore invisible to data contributors. In the MATLAB codebase, TAXCODE appears in FieldDefinitions as a standard field, but no rule module validates consistency between TAXCODE and SPEC-CODE; the two are kept consistent via the database import process.

NUMBER

Type	double
NULL	Allowed
Lookup	—
Surveys	All (sighting records)

Best estimate of the number of animals in the sighting group. Should be $\geq$ NUMCALF. This constraint is not currently enforced in code.

NUMCALF

Type	double
NULL	Allowed
Lookup	—
Surveys	All (sighting records)

Number of calves observed within the sighting group. Must be $\leq$ NUMBER (not enforced in code). Both zero and NULL are acceptable when no calves were observed.

3.5 Sighting Characteristics

BEHAV1–BEHAV15

Type	double (each field)
NULL	Allowed for BEHAV2–15; BEHAV1 is expected when behavior is observed

Lookup	data/tables/Behave.csv (90 rows)
Surveys	All (sighting records)

Up to 15 simultaneous behavioral observation codes per sighting. Slots are filled left to right; unused slots are NULL. No duplicate codes within a single sighting are expected. `behavioral_rules.m` validates all populated BEHAV slots against the 90 codes in Behave.csv.

Behavior codes cover general locomotion states (traveling, milling, logging, breaching), feeding behaviors, social behaviors, injury/mortality indicators, and vessel interaction codes. Codes 73 (actively fishing) and 74 (actively fishing with deployed gear) were added in Version 8 for vessel sightings (TAXCODE=0).

The following synthetic variables are derived from BEHAV_n fields by SAS macros and are not stored in the database: CALF, DEAD, FEED, FISHING, GEAR, HURT, JELL, MILL, POOP, SAG, STRK, WAKE, WHLR.

ANHEAD

Type	double
NULL	Allowed
Lookup	data/tables/ANHEAD.csv (20 rows)
Surveys	Aerial and some shipboard (sightings)

Coded compass heading of the observed animal's head at the time of sighting, recorded as a directional interval rather than an exact bearing. The 20 codes represent compass segments; code 20 (added in Version 8) indicates 360° milling or circling with no consistent heading. Used in computing the synthetic variable MILL.

IDREL

Type	double
NULL	Allowed
Lookup	data/tables/IDREL.csv (4 rows)
Surveys	All

Identification reliability code (4 levels: sure/positive, probable, possible, not recorded). Used to generate the synthetic variable ID during data delivery.

CONFIDNC

Type	double
NULL	Allowed
Lookup	data/tables/Confidnc.csv (12 rows)
Surveys	All

Overall sighting confidence level.

PHOTOS

Type	double
NULL	Allowed
Lookup	data/tables/PHOTOS.csv (5 rows)
Surveys	All (sighting records)

Code indicating the type of photographic record obtained for the sighting (e.g., none, still photography, video, aerial imagery). A `photos_rules.m` module exists in the rules directory but is not confirmed as wired into `SurveyValidator.m`.

3.6 Platform

PLATFORM

Type	double
NULL	Allowed
Lookup	data/tables/PLATFORM.csv (283 rows)
Surveys	All

Numeric code identifying the survey vessel or aircraft. The code space is divided into named ranges:

Range	Category
001–019	USCG vessels
020–099	NOAA and federal research vessels
100–199	Foreign research and institutional vessels

Range	Category
200–299	Sailing vessels and whale-watch boats
300–349	CETAP charter vessels
350–374	Canadian and commercial platforms
375–424	Charter fishing vessels
425–474	Passenger ferries
475–525	Tugs and work boats
526–550	BLM charter vessels
551–599	Miscellaneous vessels
600–619	Helicopters
620–625	Private aircraft
626–639	Dedicated survey aircraft
640–644	U.S. Coast Guard aircraft
645–699	Miscellaneous aircraft
700–999	Other (shore stations, internet sources, stranding data, literature, etc.)

`platform_rules.m` validates PLATFORM values against PLATFORM.csv.

3.7 Environmental Conditions

BEAUFORT

Type	double
NULL	Allowed
Lookup	data/tables/Beaufort.csv (13 rows)
Surveys	All (required for on-watch records in aerial and shipboard)

Beaufort sea state (0–12). Beaufort.csv covers the full 0–12 scale (13 entries).

The description in `FieldDefinitions` (“Beaufort sea state (0-9)”) is incorrect; see Section 6.3. `BEAUFORT` and `VISIBLTY` together define acceptable survey effort; the standard on-effort threshold is `Beaufort ≤ 3` and `VISIBLTY ≥ 2` n.mi. `beaufort_rules.m` validates against the lookup table.

VISIBLTY

Type	double
NULL	Allowed
Lookup	—
Surveys	All (required for on-watch records in aerial and shipboard)

Estimated clear visibility in nautical miles, to the nearest 0.1 n.mi. Required for all on-watch records in aerial and shipboard data; optional elsewhere.

Legacy negative values persist in the archived database (migrated from the old `OLDVIZ` field during the 2020 update): `-1` = clear ≥ 2 n.mi., `-2` = < 2 n.mi. fog, `-3` = < 2 n.mi. haze, `-4` = < 2 n.mi. rain, `-5` = < 2 n.mi. snow. No new data should use negative `VISIBLTY` values. `environmental_rules.m` currently permits negative values (`visibility_allow_negative = true`) but carries a `FIXME` indicating that this should be restricted.

SURFTEMP

Type	double
NULL	Allowed
Lookup	—
Surveys	All (optional)

Sea surface temperature in degrees Celsius. Optional for all record types. Formerly called `WTEMP` during CETAP. Can be measured by airborne radiometer (aerial surveys) or shipboard instruments. SST data from airborne radiometers have been erratic in some datasets; users should apply caution. `environmental_rules.m` issues warnings for values outside -2 to 35°C .

CLOUD

Type	double
NULL	Allowed
Lookup	<code>data/tables/Cloud.csv</code> (6 rows)
Surveys	All (optional)

Cloud cover in oktas (eighths of sky covered). The standard scale is 0–8. The description in FieldDefinitions (“Cloud cover (0-10)”) is incorrect; see Section 6.3. Cloud.csv has only 6 rows, which may indicate an incomplete or stale snapshot (see Section 6.4). Cloud cover validation has been commented out in `environmental_rules.m` pending review.

WX

Type	string
NULL	Allowed
Lookup	data/tables/WX.csv (12 rows)
Surveys	All (optional; strongly encouraged)

One-character weather condition code. Added in 2004 alongside the re-designed VISIBLTY field. Valid values:

Code	Condition
B	Rain (or other precipitation) and fog
C	Clear
D	Drizzle
F	Fog
G	Gray (heavy overcast, no precipitation)
H	Haze
L	Light rain / intermittent showers
P	Patchy fog
R	Rain
S	Snow
T	Thunderstorms / squalls
X	Not recorded

WX validation was implemented in `environmental_rules.m` but has been commented out with a FIXME noting that validation should be done via the lookup table through `foreign_key_rules.m`.

GLAREL

Type	double
NULL	Allowed
Lookup	data/tables/GLARE.csv (4 rows)
Surveys	All (optional)

Glare severity on the left (port) side of the platform (0=none, 1=light, 2=moderate, 3=severe).

GLARER

Type	double
NULL	Allowed
Lookup	data/tables/GLARE.csv (4 rows)
Surveys	All (optional)

Glare severity on the right (starboard) side. Same scale as GLAREL.

4. Cross-Field Rules

4.1 Rules Enforced in Code

The following rules are checked by validation modules in `src/+narwc/+validation/+rules/`. `SurveyValidator.m` orchestrates all modules.

Rule	Fields involved	Module	Severity
LAT_DD and LONG_DD must both be present or both absent	LAT_DD, LONG_DD	coordinate_rules.m	error
LAT_DD must be in range -90 to 90	LAT_DD	coordinate_rules.m	error
LONG_DD must be in range -180 to 180	LONG_DD	coordinate_rules.m	error
Coordinates outside configured survey area (default 20-55°N, -85 to -40°W)	LAT_DD, LONG_DD	coordinate_rules.m	warning
YEAR in range 1900 to current year + 1	YEAR	datetime_rules.m	error
YEAR before 1980	YEAR	datetime_rules.m	warning
MONTH must be 1-12	MONTH	datetime_rules.m	error
DAY must be 1-31	DAY	datetime_rules.m	error
YEAR/MONTH/DAY must form a valid calendar date	YEAR, MONTH, DAY	datetime_rules.m	error
TIME must be valid HHMMSS (HH < 24, MM < 60, SS < 60)	TIME	datetime_rules.m	error
BEAUFORT must be in Beaufort.csv	BEAUFORT	beaufort_rules.m	error
BEHAV1-15 codes must be in Behave.csv	BEHAV1-15	behavioral_rules.m	error
SPECCODE must be in SPECCODE.csv	SPECCODE	species_rules.m	error
PLATFORM must be in PLATFORM.csv	PLATFORM	platform_rules.m	error
VISIBLTY must be ≥ 0 (configuration dependent; currently allow_negative=true)	VISIBLTY	environmental_rules.m	error
VISIBLTY > 50 n.mi. is unusually high	VISIBLTY	environmental_variables.m	warning
SURFTEMP outside -2 to 35°C	SURFTEMP	environmental_variables.m	warning

Rule	Fields involved	Module	Severity
Required fields (currently LAT_DD, LONG_DD, YEAR, MONTH, DAY) non-NULL	multiple	required_fields.m	error

The required-fields list is a genuinely open question, not settled fact — see PROJECT_STATUS.md §7 (“required_fields.m accuracy”). required_fields.m’s own fallback list (used when no config.required_fields override is supplied) is {'LAT_DD', 'LONG_DD', 'YEAR', 'MONTH', 'DAY'}, matching validation_config_default.m. This is a minimal baseline, not necessarily correct against the database schema’s actual NOT NULL constraints (see database_schema.md’s Master Table section — only FILEID/EVENTNO are NOT NULL at the column level). Do not assume this list is final without checking the current code.

4.2 Rules Documented in PDF but Not Enforced in Code

Rule	Fields	Assessment
SIGHTNO required for sightings; absent for non-sightings	SIGHTNO	Possibly unimplemented
SPECCODE required for sightings; absent for non-sightings	SPECCODE	Possibly unimplemented
SIGHTNO must be unique within a file	SIGHTNO, FILEID	Possibly unimplemented
STRIP required when LEGTYPE=2 and LEGSTAGE=2; not allowed otherwise	STRIP, LEGTYPE, LEGSTAGE	Possibly unimplemented
ANGLEL/ANGLER replace STRIP for NEAQ 2022+; not mixed with STRIP	ANGLEL, ANGLER, STRIP	No mutual-exclusion rule
TAXCODE must be consistent with SPECCODE	TAXCODE, SPECCODE	Likely intentional — TAXCODE is DB-assigned
NUMBER must be $\geq$ NUMCALF	NUMBER, NUMCALF	Possibly unimplemented
S_LAT and S_LONG must both be present or both absent	S_LAT, S_LONG	Possibly unimplemented
ALT should be within reasonable range for survey aircraft	ALT	Possibly unimplemented
HEADING must be 0–360°	HEADING	Possibly unimplemented

Rule	Fields	Assessment
LEGNO is sequential within a file	LEGNO, FILEID	Likely intentional — gaps are permitted

5. Lookup Table Inventory

All files are in data/tables/. They are CSV snapshots exported from the production SQL Server database via scripts/setup/pull_lookup_tables.m. The production database is authoritative; these snapshots may drift from the live database as codes are added or retired.

File	Rows	Constrains field(s)	Notes
ANHEAD.csv	20	ANHEAD	Codes 1–20; code 20 (milling) added in Version 8
Beaufort.csv	13	BEAUFORT	Full scale 0–12; FieldDefinitions description of 0–9 is incorrect
Behave.csv	90	BEHAV1– BEHAV15	90 distinct behavior codes; codes 73–74 added in Version 8
Block.csv	55	BLOCK	Survey block identifiers
Cloud.csv	6	CLOUD	Fewer than 9 rows expected for full 0–8 okta scale; possibly stale
Confidnc.csv	12	CONFIDNC	Sighting confidence levels
Contrib.csv	23	—	Data contributor lookup; not a direct field FK constraint
DDSOURCE.csv	18	DDSOURCE	Data delivery source codes
DType.csv	5	—	Survey type codes; fifth code not fully documented in PDF
GLARE.csv	4	GLAREL, GLARER	Values 0–3
IDREL.csv	4	IDREL	Identification reliability (sure, probable, possible, not recorded)
IDSOURCE.csv	53	IDSOURCE	Identification source codes
LEGGOOD.csv	2	—	Leg usability flag (2 values)
LEGSTAGE.csv	3	LEGSTAGE	Stage-within-leg codes
LEGTYPE.csv	9	LEGTYPE	Leg-type codes
MONTH.csv	16	MONTH	16 rows for 12 months; extra rows likely display metadata
OLDVIZ.csv	5	—	Legacy visibility codes –1 to –5; field retired; for reference only
PHOTOS.csv	5	PHOTOS	Photo type codes

File	Rows	Constrains field(s)	Notes
PLATFORM283.csv	283	PLATFORM	Full platform enumeration consistent with PDF
SPECCODE315.csv	315	SPECCODE	All active codes across all TAXCODE groups including TAXCODE=0 and birds
STRATUM.csv	10	STRATUM	X, Y, Z, 0, A, B, I, O, M, R
STRIP.csv	16	STRIP	Distance interval codes
sysdiagrams0.csv	—	—	SQL Server system table artifact; no validation use
TAXCODE.csv	10	TAXCODE	Values 0–9
WX.csv	12	WX	Weather codes B, C, D, F, G, H, L, P, R, S, T, X

6. PDF–Code Gaps

6.1 Likely Deprecated (CETAP-era or formally retired)

Item	Description
Negative VISIBLTY values (–1 to –5)	Migrated from the retired OLDVIZ field during the 2020 database update. Exist in the archive for historical records only. The PDF explicitly prohibits their use in new submissions.
OLDVIZ.csv	Retained for reference; the OLDVIZ field has been retired and values folded into VISIBLTY.
HUMANACT and DEBRIS fields	CETAP-era codes for human activities and debris. Replaced by TAXCODE=0 sightings with SPECCODE values.
Italicized TAX-CODE=0 SPEC-CODEs	AC-J, AC-P, AC-S, AC-T, BT-H, BT-L, BT-M, DIVE, DU-G, DU-T, MULT, SONR, SWIM are obsolete CETAP HUMANACT codes explicitly prohibited in the NARWC database.
SIGHTNO=0 for non-sightings	dBASE submission artifact; converted to NULL during archival.
Four-digit TIME (hhmm)	No longer accepted for survey data; expanded to HHMMSS by appending “00”.

6.2 Possibly Unimplemented

Item	Fields	Description
Sighting-record constraints STRIP conditional requirement	SIGHTNO, SPEC-CODE, STRIP, LEG-TYPE, LEGSTAGE	Both fields have required/prohibited conditions based on whether a record is a sighting, but no rule module implements these checks. The PDF requires STRIP for on-effort (LEGTTYPE=2, LEGSTAGE=2) animal sightings. Not enforced.
ANGLEL/ANGLE vs. STRIP exclusivity	ANGLEL, ANGLER, STRIP	The two systems are mutually exclusive by survey program but not enforced in code.
NUMBER ≥ NUMCALF	NUMBER, NUM-CALF	Logical constraint; not enforced.
S_LAT/S_LONGS co-presence	S_LAT, S_LONG	Should follow the same both-present-or-both-absent rule as LAT_DD/LONG_DD; not implemented.
SIGHTNO uniqueness within file	SIGHTNO, FILEID	Duplicate detection not implemented.
VISIBLTY negative value restriction	VISIBLTY	environmental_rules.m has visibility_allow_negative = true with a FIXME noting this should be restricted to legacy records only.

6.3 Known Errors in FieldDefinitions

Field	FieldDefinitions value	Correct value	Source
BEAUFORT description	“Beaufort sea state (0-9)”	Range is 0–12	PDF §8.A.2; Beaufort.csv has 13 rows
CLOUD description	“Cloud cover (0-10)”	Scale is 0–8 oktas	PDF §8.A.4; standard okta scale
ALT description	“Altitude in meters”	Values are in feet	PDF §8.A.1

6.4 Unclear

Item	Description
MONTH (16 rows)	The table has 16 rows; calendar months should yield 12. Extra rows may be display metadata or schema artifacts. No impact on validation since MONTH is validated by range check in <code>datetime_rules.m</code> , not by lookup.
Cloud.csv (6 rows)	A complete 0–8 okta scale requires 9 rows. The 6-row table may be incomplete. Cloud validation is commented out, so this does not currently affect data quality checks.
DType.csv (5th row)	The reference PDF describes four survey type codes (A, F, H, O). The fifth entry in DType.csv is undocumented in the PDF content reviewed.

7. Deprecated / Legacy Fields

The following fields are described in the reference PDF (Chapter 8, Part C: “Variables that no longer exist”) or are otherwise known to be retired. They are absent from `narwc.db.FieldDefinitions` and should not appear in new data submissions.

Field	Former description	Replaced by
ADEPTH	Meter depth (meters) at exact sighting position. CETAP line-transect aerial only.	No direct replacement
ALATDEG	Degrees of latitude at exact sighting position. CETAP line-transect aerial only.	S_LAT
ALATMIN	Minutes of latitude at exact sighting position. CETAP only.	S_LAT
ALATSEC	Seconds of latitude at exact sighting position. CETAP only.	S_LAT
ALONGDEG	Degrees of longitude at exact sighting position. CETAP only.	S_LONG
ALONGMIN	Minutes of longitude at exact sighting position. CETAP only.	S_LONG
ALONGSEC	Seconds of longitude at exact sighting position. CETAP only.	S_LONG
CETSPRCD	Legacy dBASE name for the species code field.	SPECCODE
DEBRIS	CETAP code for debris/pollution observations.	SPECCODE with TAXCODE=0
HUMANACT	CETAP code for human activities observed.	SPECCODE with TAXCODE=0

Field	Former description	Replaced by
OLDVIZ	Legacy one-digit visibility condition code (values -1 to -5).	VISIBLTY (negative values, historical records only)
WTEMP	CETAP name for sea-surface temperature.	SURFTEMP

Additional retired fields are documented in Part C of the reference PDF. The fields above are those with confirmed CETAP provenance or direct successors in the current schema.

7.1 Synthetic Variables (Part B)

The following variables are computed on demand from database fields by SAS macros and are not stored in the database. They appear in data deliveries but must not be submitted as input data: CALF, CANADA, DEAD, FEED, FISHING, GEAR, HURT, ID, JDATE, JELL, MILL, POOP, SAG, SEASON, SPECNAME, STRK, TAXTYPE, TYPE, WAKE, WHLR.